

Yurt 8 is one system that watches every project in the portfolio, generates the next design move on demand, and learns from both, so each pass gets faster, cheaper, and sharper. It is built so leadership sees dollars and time move every single month, and it is built to Hut 8's mission: scaling AI, and using AI to scale itself. Every element below passed five tests or got cut:

Keep it simple

Cut cost

Cut time

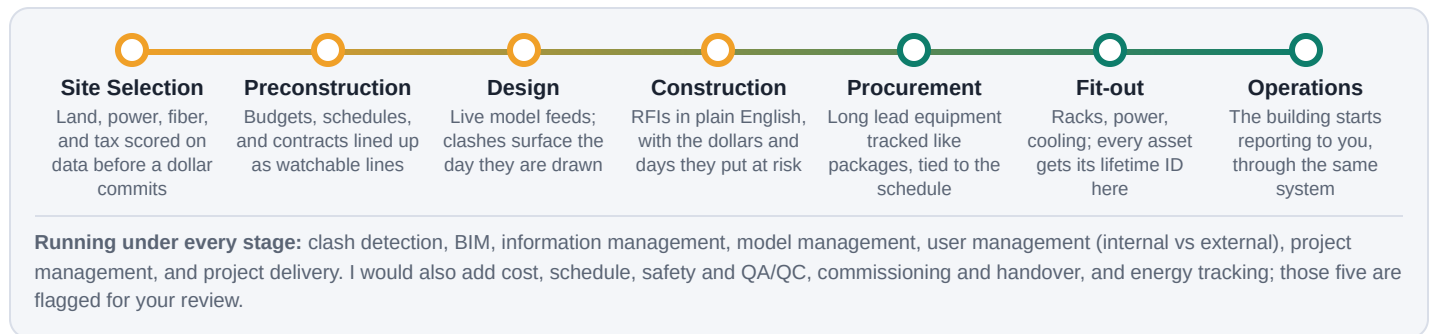
Raise quality

Built to scale, never stagnant

THE FACE

A pizza tracker for buildings

Domino's earned trust by letting people watch the pizza move. Owners deserve the same view of their capital projects. Every Hut 8 project lives on one chronological tracker, and every stage is wired to the live data that matters for it. Not a status slide. The live state of the work.



THE ENGINE

Review. Generate. Learn. Repeat.

Review

See everything as it happens, not at the monthly meeting.

- Live feeds of working Revit models, CAD files, and standardized spreadsheets
- RFIs, submittals, and financials translated into plain English for the owner
- Clashes and design issues surfaced as they occur, before they become change orders
- Connects to ACC as the main hub, plus contractors' Procore, Tableau, Power BI, and Microsoft Project

Generate

Design moves in seconds instead of weeks, from my proven generative work on live programs.

- Deterministic guardrails alternate with AI generators, gate after gate, so output is fast and safe
- Test fits and sizing with no third party fees for work that needs no stamp
- Runs in a browser through Autodesk's cloud; nobody needs Revit installed
- Swap cabinet vendors, re-size MEP and containment, re-wire redundant cooling, swap GPUs; every asset number is preserved

Learn

Every pass makes the next one better. The system never stands still.

- AWS Bedrock agents on S3 data lakes that grow with every review and generation
- Knowledge gated by review before it teaches anything
- Tied to the codes, standards, and rules Hut 8 and its vendors flag as critical
- Grows from stakeholder input and exemplar designs fed in as standardized JSON

One shared language: Review, Generate, and Learn all read and write the exact same JSON format { *coordinates*, *parameters*, *relationships*, *patterns* }. Nothing gets translated between steps, so nothing gets lost between steps.

ACC, kept deliberately simple

Vendors do day to day design in Autodesk Construction Cloud, simplified through clean permissions. For Hut 8 it is a limited tool on purpose: a data source we mine by API. Owners do not need forty modules. They need the tracker.

ISO 19650 without the ceremony

HUT8-[PROJECT]
 ↳ 01-WIP each vendor's own working area
 ↳ 02-SHARED coordinated, clash reviewed
 ↳ 03-PUBLISHED approved to build, read only
 ↳ 04-ARCHIVE every version, kept automatically

Assets that report for life

Every asset gets one ID at design and keeps it forever: in the model, on the physical tag, in the sensor feed. IoT site monitoring flows into Operations through the same JSON. The over-engineered version of this standard that never survives an American job site is exactly what this is not.

THE PLAN

Twenty four months, reported monthly in dollars and days

Build the tool, build the team, then maintain and scale both. Every month closes with one page for leadership: dollars moved, days moved, and which scenario the data came from, alpha, beta, or live. If a month cannot show both numbers, that month failed and I will say so plainly.

MONTHS 1 TO 2

Wire into what exists

ACC mining live on one active project. Tracker alpha stands up. First plain English feeds reach the owner. Existing contractors and contracts keep their tools; Yurt 8 conforms to the work.

Checkpoint: first clash caught on screen, priced against the change order it replaced. Source: live.

MONTHS 3 TO 4

Review, fully live

Model feeds, financials, and clash surfacing across the active portfolio. Procure and schedule connections land.

Checkpoint: avoidable slowdowns identified and priced, monthly. Source: live.

MONTHS 5 TO 6

Generate, first wins

Test fits and sizing on the real cabinet library. Work that took weeks and third party fees now takes seconds.

Checkpoint: third party design fees eliminated, itemized per run. Source: beta on live layouts.

MONTHS 7 TO 9

Second renaissance engineer

Generate moves into the browser; anyone Hut 8 authorizes can run it without Revit. Hire two arrives.

Checkpoint: design cycle time, before vs after, in days. Source: live.

MONTHS 10 TO 12

Learn, gated and growing

Knowledge bases go live, gated by review, tied to the codes and standards Hut 8 flags. Year one closes with a full tally.

Checkpoint: year one savings tally, line by line. Source: live plus beta.

MONTHS 13 TO 18

Portfolio wide, third engineer

Every active project on the tracker. IoT monitoring lights up Operations. New projects start with generative site selection from day one.

Checkpoint: change order rate vs baseline, portfolio wide. Source: live.

MONTHS 19 TO 24

Full lifecycle, compounding

Site selection through operations on one loop. Every review sharpens generation; every generation feeds the lakes.

Checkpoint: year two tally, and year three priced from real data. Source: live.

\$350K

Year one ask, all in

\$4.0M

Year one savings, base case

11x

Year one return

\$10M

Year two savings

LINE	YEAR 1	YEAR 2
What it costs		
Me, all in	\$250K	\$350K
Engineers two and three	\$0	\$520K
Licensing (ACC, APS)	\$45K	\$90K
Cloud and API tokens	\$55K	\$90K
Total cost	\$350K	\$1.05M
What it returns		
Change orders avoided	\$2.4M	\$6.0M
Third party design fees eliminated	\$0.9M	\$1.8M
Schedule value captured	\$0.7M	\$2.2M
Total savings, base case	\$4.0M	\$10.0M

Why these savings are real at Hut 8's scale

Hut 8 has 830 MW under construction at its guided \$9M to \$11M per MW: roughly \$8B of construction value in flight, backed by \$7.5B of project bonds closed in 2026. Studies put change orders from design errors alone at 3 to 5% of construction budget, and documented savings from BIM clash detection reach 10% of contract value. Year one claims 0.75% of a \$400M watched slice. Year two claims 0.5% of wider coverage as River Bend and Beacon Point halls deliver. A fraction of the documented ceiling, on purpose.

On meetings: meetings are for decisions. Status lives on the tracker. No decision, no meeting.

On the team: one, then three. Every hire a renaissance engineer who rejects role boundaries. Critical thought, delivered results, first principles, real teamwork. No seat fillers, ever. The system prevents chaos; the team stays drilled for forest fires anyway.

Show my homework: the interactive demo opens the hood: the exact prompt that drove my AI build agents, the architecture, and the sources. I built this in two days with a multi agent setup I designed, and I can build the same without AI: HTML, Python, C#, C, and .NET are home turf. Point me outside my comfort zone and the same financial logic arrives; I learn fast, and I stay motivated and curious.

The 3D you will see there: web native, ships as one file, runs offline, no installs, no servers. When simulation depth earns its cost, the path is NVIDIA Omniverse and the open source OpenUSD ecosystem.

Sources: Hut 8 Q1 2026 results (May 6, 2026); FY2025 Form 10-K; Beacon Point release (July 20, 2026); Vega energization (June 30, 2025); JLL 2025 data center cost index; AIA change order study; Stanford CIFE BIM benchmark; Navigant rework and RFI studies. Full list with links: github.com/PracticalConstructs/hut8-casestudy.